MIL-V-23151B(SH)
AMENDMENT 2
20 October 1978
SUPERSEDING
AMENDMENT 1
17 June 1968

MILITARY SPECIFICATION

VOLTMETER, EXPANDED SCALE SWITCHBOARD TYPE

(NAVAL SHIPBOARD USE)

This amendment forms a part of Military Specification MIL-V-23151B(SHIPS), dated 10 April 1967.

PAGE 1

1.1, 1.2 and 1.2.1: Delete and substitute:

"1.1 Scope. This specification covers high accuracy, shockproof switchboard mounted, expanded scale voltmeters.

"1.2 Classification.

"1.2.1 Type designation. The type designation of voltmeters shall be in the following form, and as specified (see table II and 6.2):

MR	49	W	115	AF	H
Component (1.2.1.1)	Style (1.2.1.2)	Color scheme (1.2.1.3)	Center-scale value (1.2.1.4)	Kind of current (1.2.1.5)	Shockproof feature (1.2.1.6) "

PAGE 2

1.2.1.4: Add the following:

"Center-scale value	Scale range
115	110-120
120	115-125
208	198-218
220	210-230
440	420-460
450	430-470 "

3.4.2.1, line 2: Delete "No. 11" and substitute "No. 111".

3.4.3: Delete and substitute:

"3.4.3 Expanded scale circuit. The expanded scale circuit components shall be encapsulated in material which shall cover all components and shall be in contact with the interior of the meter barrel extension or other case or container of the circuit."

PAGE 4

3.6: Delete "3.22" and substitute "3.21".

PAGE 5

3.15: Delete and substitute:

"3.15 Shock (high-impact). When voltmeters are tested in accordance with 4.3.14 no screws, bearings, pivots, windows, or other parts shall become loosened, cracked, (except as specified herein), or unduly damaged. Minor cracking, or distortion of pivots will be permissible only when conformance to the post shock performance has been demonstrated.

FSC 6625

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:43Z
Check the source to verify that this is the current version before use.

Cracking of glass windows will be permissible only when readability of the complete scale is retained and no fragment has become dislodged. The permanent change in indication based on calibration made after resetting the pointer to zero shall not exceed the value specified (see table II)."

PAGE 10

5.1: Add "(see 5.3)."

PAGE 11

5.2.1.1 and 5.2.2: Add "(see 5.3)."

Add the following paragraphs:

"5.3 Cushioning, filler, dunnage, and wrapping materials.

"5.3.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill materials for packaging and packing applications such as cushioning, filler, or dunnage is prohibited for materials destined for shipboard installation/stowage.

"5.3.2 Level C preservation-packaging and packing. When loose-fill type materials are used for packaging and packing applications such as cushioning, filler, and dunnage, all containers (unit, intermediate, and shipping) shall be marked or labelled with the following information:

'CAUTION

Contents cushioned with loose-fill material.
Not to be taken aboard ship.
Remove and discard loose-fill material.
If required, recushion with cellulosic material, bound fiber, fiberboard, or transparent flexible cellular material.'

"5.3.3 Cushioning, filler, dunnage, and wrapping materials selected, whenever available, shall exhibit improved performance for resistance to fire."

NOTE: The margins of this amendment are marked "#" to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
Navy - SH
(Project 6625-N740)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:43Z
Check the source to verify that this is the current version before use.